

12.2 The significance of artificial insemination for estimation of breeding values

Naturally artificial insemination (AI) has had great practical significance in the breeding of farm animals. By means of AI it is possible to utilize elite animals to a much greater extent than by natural mating. But AI really comes into its own and shows its significance as estimation of breeding values becomes independent of the ownership. Who owns the daughters of an elite bull? When using AI no single person can own all daughters of an AI bull, see also section 12.1 for rules of estimation of the breeding value. The most important factors when estimation breeding values are accuracy and impartiality.

The most important factors for estimation of breeding value

1. The estimated breeding values must be as accurate as possible
2. The estimated breeding values must be independent of ownership
3. The estimated breeding values of all candidates must be comparable

From point 1 it becomes clear that AI increases the size of the family and thereby also the accuracy of the estimated breeding value. In order to comply with the conditions in point 2 and 3 it was formerly common practice to ship the animal to a test station. This ensured that the results were independent of the ownership. At the same time the results were comparable, since they were obtained in the same environment.

Admittedly, test stations do have great problems with point 1. It is both difficult and expensive to place a large number of animals on a test station. The biggest problem concerning the test stations is the definition of their environmental condition. Which environment is most appropriate to ensure that the selected animals are adapted to the future production system? This question has no proper answer.

AI can solve the problems concerning point 2 and 3. Point 1 is at the same time dealt with in the best possible way. Extensive AI and adequate recording can ensure that the results obtained from the production herds can be used in estimation of the breeding values. This means that the selection takes place in a place which closely resembles the present production environment. Thus the collection of field data is much more advantageous than the former test stations for swine and dairy. The use of open recording system is only possible when extensive AI is practised. Since the offspring of several different bulls are present in a herd, it is possible to utilize information from each animal as deviations from the average of the herd, instead of using the absolute data. This is quite important, as a bull will not necessarily get a fair evaluation, since it is not used evenly in low or high production herds.

Test stations are still necessary in cases where AI has not been fully developed and in cases where the owner can influence the results in one way or another. Test stations can also be of some value in the measuring of characteristics, which is impossible to record in the practical production.

For cattle and swine the AI is so well developed that almost all former test stations have been closed.

For hen, mink and rainbow trout test stations is still needed for comparable results.

The reproduction in dog and horse breeding is almost similar to the way AI is practised, even though most of these matings are natural. Here a special problem occurs though, a tendency to mate high rank animals or low rank animals. This problem can be overcome by working with a full Animal Model. Random mating is required if a simple model is used.

Other conditions can ruin the estimated breeding value, for instance partiality, therefore a good or bad result should not influence the recordings. When hip dysplasia in a dog is scored at a local X-ray clinic, the owner might not want to pass a bad score on to the Kennel Club, whereas he would very much like to record a good score. It is difficult to compensate for such a bias in relation to the recordings, even when very sophisticated statistical methods are utilized.